

Practical 4

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4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the `groupby` and `mean()` operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

- The user should enter a list of numbers separated by space when prompted.

Output Format:

- The program should display the mean of even and odd numbers separately.
- Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Refer to the sample test cases for better understanding regarding input and output format.

Sample Test Cases

seriesMa...

```
1 import pandas as pd
2
3 # Take inputs from the user to create a list of numbers
4 numbers = list(map(int, input().split()))
5
6 # Create a Pandas series from the list of numbers
7 s = pd.Series(numbers)
8
9 # Grouping by even and odd numbers and calculating the mean
10 grouped = s.groupby(s % 2 == 0).mean()
11
12 # Display the mean of even and odd numbers with labels
13 grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]
14 print("Mean of even and odd numbers:")
15 print(grouped)
```

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4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

- Convert the dictionary to a Pandas DataFrame.

Add a new row:

- Take inputs from the user for the new row data (name, age).
- Add the new row to the DataFrame.
- Display the DataFrame after adding the new row.

Modify a row:

- Modify a specific row by changing the age. Take the row index and new age value from the user.
- Display the DataFrame after modifying the row.

Delete a row:

- Take the row index to be deleted from the user.
- Remove the specified row.
- Display the DataFrame after deleting the row.

Add a new column:

- Add a column **Gender** with values taken from the user.
- Display the DataFrame after adding the new column.

Modify a column:

- Convert names to uppercase.
- Display the DataFrame after modifying the column.

Delete a column:

- Remove the **Age** column.
- Display the DataFrame after deleting the column.

Sample Test Cases

datafram...

```
1 import pandas as pd
2
3 # Provided dictionary of lists
4 data = {
5     'Name': ['Alice', 'Bob', 'Charlie'],
6     'Age': [25, 30, 35],
7 }
8
9 # Convert the dictionary to a DataFrame
10 df = pd.DataFrame(data)
11
12 # Display the original DataFrame
13 print("Original DataFrame:")
14 print(df)
15
16 # Adding a new row
17 name = input("New name: ")
18 age = int(input("New age: "))
19
20 new_row = {'Name': name, 'Age': age}
21 df = df.append(new_row, ignore_index = True)
22
23 # Display the DataFrame after adding a new row
24 print("After adding a row:\n", df)
25
26 # Modifying a row
27 ind = int(input("Index of row to modify: "))
28 age = int(input("New age: "))
29
30 df.loc[ind, 'Age'] = age
31
32 # Display the DataFrame after modifying a row
```

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4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

- Display the first five rows of the data frame.
- Calculate the average age of the students (limit the average age up to 2 decimal places).
- Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

Note:
Refer to the displayed test cases for better understanding.

Sample Test Cases

studentin...

```
1 import pandas as pd
2
3 # Read the text file into a DataFrame
4 file = input()
5 data = pd.read_csv(file, sep="\\s+", header=None, names=["Name", "Age", "Grade"])
6
7
8 # write your code here..
9 print("First five rows:")
10 print(data.head())
11
12 avg_age = round(data["Age"].mean(), 2)
13 print(f"Average age: {avg_age}")
14
15 print("Students with a grade up to B")
16 filtered_student = data[data["Grade"] <= "B"]
17 print(filtered_student)
```

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4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by Month and calculate the total sales for each month.
- Find the month with the highest total sales and display it.
- Also, display the total sales for the best month.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	6	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8 df["Date"] = pd.to_datetime(df["Date"])
9 df["Month"] = df["Date"].dt.strftime("%Y-%m")
10 df["Total Sales"] = df["Quantity"] * df["Price"]
11 # Find the month with the highest total sales
12 sales_by_month = df.groupby("Month")["Total Sales"].sum()
13 best_month = sales_by_month.idxmax()
14 highest_sales = sales_by_month.max()
15
16 print(f"Best month: {best_month}")
17 print(f"Total sales: ${highest_sales:.2f}")
18
```

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4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Find the product that sold the most in terms of quantity sold.
- Display the product that sold the most and the total quantity sold for that product.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	6	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 product_quantity = df.groupby("Product")["Quantity"].sum()
10
11 # Find the product with the highest total quantity sold
12 best_product = product_quantity.idxmax()
13 highest_quantity = product_quantity.max()
14
15 # Display the result
16 print(f"Best selling product: {best_product}")
17 print(f"Total quantity sold: {highest_quantity}")
18
```

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4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

- The CSV file contains the columns: Date, Product, Quantity, Price, and City.
- Group the data by City and calculate the total quantity of products sold for each city.
- Find the city that sold the most products (based on the total quantity sold).

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	6	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	5	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

Sample Test Cases

monthFor...

```
1 import pandas as pd
2
3 # Prompt the user for the file name
4 file_name = input()
5
6 # Load the data
7 df = pd.read_csv(file_name)
8
9 # write the code..
10 city_sales = df.groupby("City")["Quantity"].sum()
11 best_city = city_sales.idxmax()
12 # Display the result
13 print(f"City sold the most products: {best_city}")
14
```

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4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City.

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).

Output the product pairs that were sold most frequently.

Sample Data:

Date	Product	Quantity	Price	City
2025-01-01	Product A	5	20	New York
2025-01-01	Product B	3	15	Los Angeles
2025-01-02	Product A	7	20	New York
2025-01-02	Product C	4	30	Chicago
2025-01-03	Product B	2	15	Chicago
2025-01-03	Product A	8	20	Los Angeles
2025-01-04	Product C	6	30	New York
2025-01-04	Product B	9	15	Los Angeles
2025-01-05	Product A	3	20	Chicago
2025-01-05	Product C	10	30	Los Angeles

Explanation:

Transactions:

2025-01-01: Product A, Product B

2025-01-02: Product A, Product C

2025-01-03: Product B, Product A

2025-01-04: Product C, Product B

2025-01-05: Product A, Product C

Now, let's count how often the pairs of products appear together:

Product A and Product B appear in transactions on 2025-01-01 and 2025-01-03

Product A and Product C appear in transactions on 2025-01-02 and 2025-01-05

Product B and Product C appears in transactions on 2025-01-04

Most Frequent Product Combinations:

Product A and Product B (2 times)

Product A and Product C (2 times)

Note:

The data cannot be displayed in the file. You can refer to the sample data provided for insights.

frequent...

```
1 import pandas as pd
2 from itertools import combinations
3 from collections import Counter
4
5 # Prompt user to input the file name
6 file_name = input()
7
8 # Read data from the specified CSV file
9 df = pd.read_csv(file_name)
10
11 # write the code
12 daily_transactions = df.groupby('Date')['Product'].apply(list)
13 pair_counts = Counter()
14
15 for products in daily_transactions:
16     products = sorted(products)
17     pairs = list(combinations(products, 2))
18     pair_counts.update(pairs)
19
20 # Output the most frequent product pairs
21 if pair_counts:
22     max_count = max(pair_counts.values())
23     for pair, count in pair_counts.items():
24         if count == max_count:
25             print(f"({pair[0]} and {pair[1]}): {count} times")
26 else:
27     print("No product pairs found.")
```

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4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

1. Display the first 5 rows of the dataset.

2. Display the last 5 rows of the dataset.

3. Get the shape of the dataset (number of rows and columns).

4. Get a summary of the dataset (using .info()).

5. Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().

6. Check for missing values and display the count of missing values for each column.

7. Fill missing values in the 'Age' column with the median age.

8. Fill missing values in the 'Embarked' column with the most frequent value (mode).

9. Drop the 'Cabin' column due to many missing values.

10. Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brando, Mr. Owen Harris	male	22	1	0	A/5 21171,7,25	5		S
2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71,2833,C85,C	5		S
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7,925	5		S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53,1,C123,5	5		S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8,85	5		S
6	0	3	Noran, Mr. James	male	0	0	0	338077,8,4583,0	0		S
7	0	1	McCarthy, Mr. Timothy J	male	34	0	0	17463,51,8625,546,5	5		S
8	0	3	Paisson, Master. Gosta Leonard	male	2	3	1	349909,21,075	5		S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742,11,1333	5		S

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4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

1. Create a new column 'IsAlone' which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.

2. Convert the 'Sex' column to numeric values (male: 1, female: 0).

3. One-hot encode the 'Embarked' column, dropping the first category.

4. Get the mean age of passengers.

5. Get the median fare of passengers.

6. Get the number of passengers by class.

7. Get the number of passengers by gender.

8. Get the number of passengers by survival status.

9. Calculate the survival rate of passengers.

10. Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below.

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Brando, Mr. Owen Harris	male	22	1	0	A/5 21171,7,25	5		S
2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Thayer)	female	38	1	0	PC 17599,71,2833,C85,C	5		S
3	1	3	Heikkinen, Miss. Laina	female	26	0	0	STON/O2. 3101282,7,925	5		S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35	1	0	113803,53,1,C123,5	5		S
5	0	3	Allen, Mr. William Henry	male	35	0	0	373450,8,85	5		S
6	0	3	Noran, Mr. James	male	0	0	0	338077,8,4583,0	0		S
7	0	1	McCarthy, Mr. Timothy J	male	34	0	0	17463,51,8625,546,5	5		S
8	0	3	Paisson, Master. Gosta Leonard	male	2	3	1	349909,21,075	5		S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27	0	2	347742,11,1333	5		S
10	1	2	Nasser, Mrs. Nicholas (Adele Achen)	female	14	1	0	237736,30,0708,C	0		S

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4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Calculate the survival rate by class.
- Calculate the survival rate by embarkation location (Embarked_S).
- Calculate the survival rate by family size (FamilySize).
- Calculate the survival rate by being alone (IsAlone).
- Get the average fare by passenger class (Pclass).
- Get the average age by passenger class (Pclass).
- Get the average age by survival status (Survived).
- Get the average fare by survival status (Survived).
- Get the number of survivors by class (Pclass).
- Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below.

Passenger id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,"Braund, Mr. Owen Harris",male,22,1,0,A/5 2137,7.25,S
1,1,"Guthrie, Mrs. John Bradley (Florence Briggs Thayer)",female,36,1,0,PC 17599,71.0033,C85,C
1,1,"Heikkinen, Miss. Laina",female,26,0,0,STON/O 3101282,7.92,S
1,1,"Payralde, Mrs. Jacques Heath (Lily May Peel)",female,19,1,0,13300,53.1,C323,S
1,0,"Allen, Mr. William Henry",male,35,0,0,17463,0.8,S
1,0,"Moran, Mr. James",male,0,0,0,10837,0.463,Q
1,0,"McCarthy, Mr. Timothy J",male,34,0,0,11041,51.8625,46,S
1,0,"Palsson, Master. Gosta Leonard",male,1,1,1,10400,21.875,S
1,1,"Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg)",female,27,0,2,36742,33.3333,S
1,1,"Nasser, Mrs. Nicholas (Adelie Aches)",female,34,1,0,10773,10.4708,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

TitanicDataset...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data['FamilySize'] = data['SibSp'] + data['Parch']
7 data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1)
8 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
9
10 # 1. Calculate the survival rate by class
11 print(data.groupby('Pclass')['Survived'].mean())
12
13 # 2. Calculate the survival rate by embarked location
14 print(data.groupby('Embarked_S')['Survived'].mean())
15
16 #3. Calculate the survival rate by family size
17
18 print(data.groupby('FamilySize')['Survived'].mean())
19
20 #4. Calculate the survival rate by being alone
21
22 print(data.groupby('IsAlone')['Survived'].mean())
23
24 #5. Get the average fare by class
25
26 print(data.groupby('Pclass')['Fare'].mean())
27
28 #6. Get the average age by class
29
30 print(data.groupby('Pclass')['Age'].mean())
31
32 #7. Get the average age by survival status
33
34 print(data.groupby('Survived')['Age'].mean())
35
36 #8. Get the average fare by survival status
37
38 print(data.groupby('Survived')['Fare'].mean())
39
40 #9. Get the number of survivors by class (sort by values descending)
41
42 print(data[data['Survived'] == 1]['Pclass'].value_counts())
43
44 #10. Get the number of non-survivors by class (sort by values descending)
45
46 print(data[data['Survived'] == 0]['Pclass'].value_counts())
47
```

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4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

- Get the number of survivors by gender (Sex).
- Get the number of non-survivors by gender (Sex).
- Get the number of survivors by embarkation location (Embarked_S).
- Get the number of non-survivors by embarkation location (Embarked_S).
- Calculate the percentage of children (Age < 18) who survived.
- Calculate the percentage of adults (Age >= 18) who survived.
- Get the median age of survivors.
- Get the median age of non-survivors.
- Get the median fare of survivors.
- Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below.

Passenger id	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,"Braund, Mr. Owen Harris",male,22,1,0,A/5 2137,7.25,S
1,1,"Guthrie, Mrs. John Bradley (Florence Briggs Thayer)",female,36,1,0,PC 17599,71.0033,C85,C
1,1,"Heikkinen, Miss. Laina",female,26,0,0,STON/O 3101282,7.92,S
1,1,"Payralde, Mrs. Jacques Heath (Lily May Peel)",female,19,1,0,13300,53.1,C323,S
1,0,"Allen, Mr. William Henry",male,35,0,0,17463,0.8,S
1,0,"Moran, Mr. James",male,0,0,0,10837,0.463,Q
1,0,"McCarthy, Mr. Timothy J",male,34,0,0,11041,51.8625,46,S
1,0,"Palsson, Master. Gosta Leonard",male,1,1,1,10400,21.875,S
1,1,"Johnson, Mrs. Oscar W (Elizabeth Vilhelmina Berg)",female,27,0,2,36742,33.3333,S
1,1,"Nasser, Mrs. Nicholas (Adelie Aches)",female,34,1,0,10773,10.4708,C
```

Note: Refer to the visible test case for better reference.

Sample Test Cases

TitanicDataset...

```
1 import pandas as pd
2 import numpy as np
3
4 # Load the Titanic dataset
5 data = pd.read_csv('Titanic-Dataset.csv')
6 data = pd.get_dummies(data, columns=['Embarked'], drop_first=True)
7
8 # 1. Get the number of survivors by gender
9 survivors_by_gender = data[data['Survived'] == 1]['Sex'].value_counts()
10 print(survivors_by_gender)
11
12 # 2. Get the number of non-survivors by gender
13 non_survivors_by_gender = data[data['Survived'] == 0]['Sex'].value_counts()
14 print(non_survivors_by_gender)
15
16 # 3. Get the number of survivors by embarked location (Embarked_S)
17 survivors_by_embarked_s = data[data['Survived'] == 1]['Embarked_S'].value_counts()
18 print(survivors_by_embarked_s)
19
20 # 4. Get the number of non-survivors by embarked location (Embarked_S)
21 non_survivors_by_embarked_s = data[data['Survived'] == 0]['Embarked_S'].value_counts()
22 print(non_survivors_by_embarked_s)
23
24 # 5. Percentage of children (Age < 18) who survived
25 children = data[data['Age'] < 18]
26 children_survival_rate = children['Survived'].mean()
27 print(children_survival_rate)
28
29 # 6. Percentage of adults (Age >= 18) who survived
30 adults = data[data['Age'] >= 18]
31 adults_survival_rate = adults['Survived'].mean()
32 print(adults_survival_rate)
33
34 # 7. Median age of survivors
35 median_age_survivors = data[data['Survived'] == 1]['Age'].median()
36 print(median_age_survivors)
37
38 # 8. Median age of non-survivors
39 median_age_non_survivors = data[data['Survived'] == 0]['Age'].median()
40 print(median_age_non_survivors)
41
42 # 9. Median fare of survivors
43 median_fare_survivors = data[data['Survived'] == 1]['Fare'].median()
44 print(median_fare_survivors)
45
```

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